

Demand Sensing Intelligence

Local demo for a Gartner-style research-planning case study.

The prototype answers two jobs:

- 1 Market scan — public news in, canonical topics, coverage of a simulated research library, ranked research actions out.
- 2 Ask the library — a client question in, a cited answer from that library out.

You should not generate documents or rebuild embeddings. The seeded SQLite library, FAISS index, and saved demo runs are in the repository. A clone only installs Python packages and Node packages, then starts the two servers. There is no FAISS build step and no “add documents” step.

A printable copy of this guide is README.pdf (for email). This Markdown file is the source of truth in Git.

Layer	URL
Frontend (use this in the browser)	http://127.0.0.1:5173
Backend API	http://127.0.0.1:8000

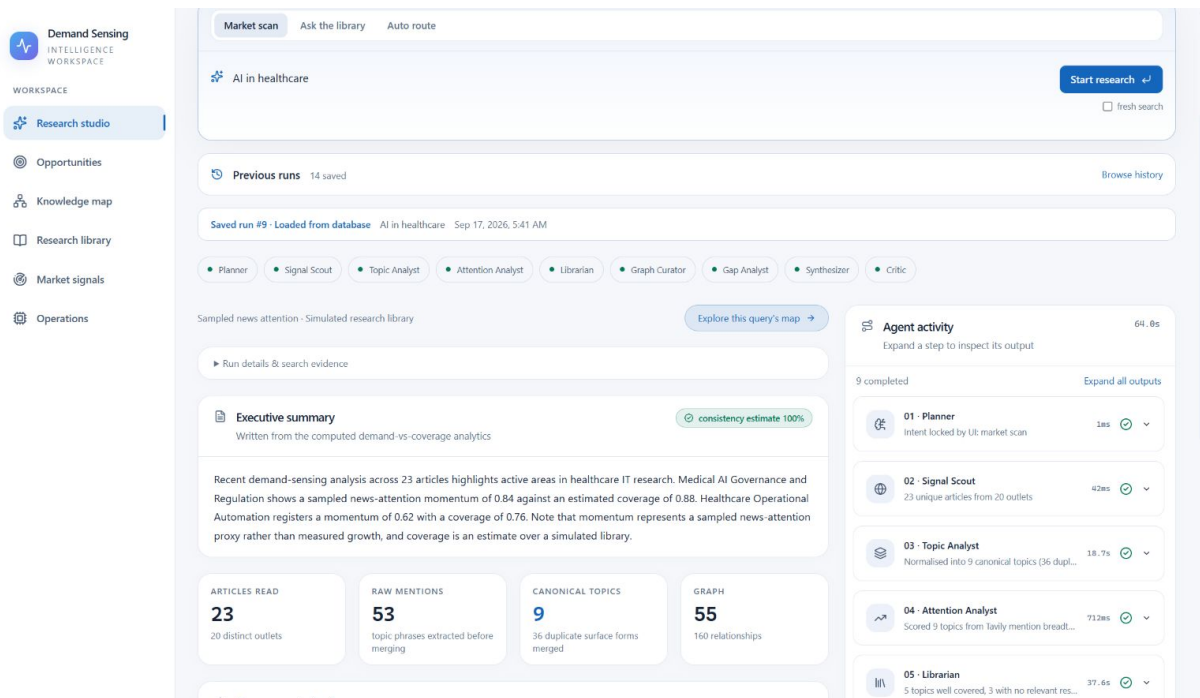
API keys (backend/.env) are not in Git. They are sent separately by email.

Product tour

These screenshots are in screenshot/ so a clone shows the workspace before anyone starts the servers.

Research studio

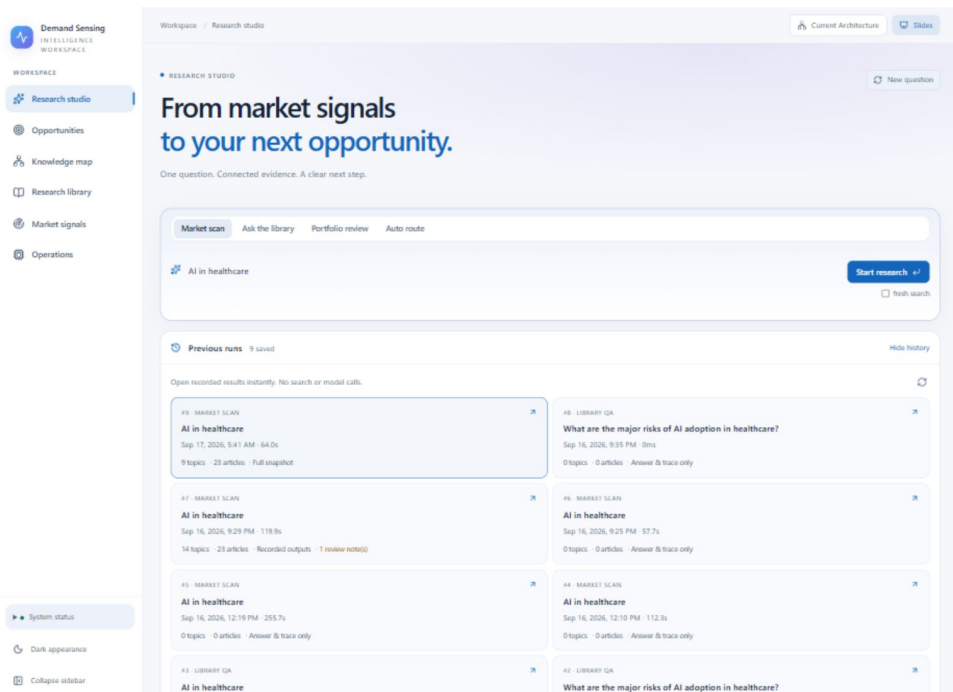
Saved Market scan for AI in healthcare (run #9 from the committed database). Nine specialists complete in order. The executive summary, article counts, and agent activity stay with the run.



Research studio showing a saved market scan for AI in healthcare

Previous runs

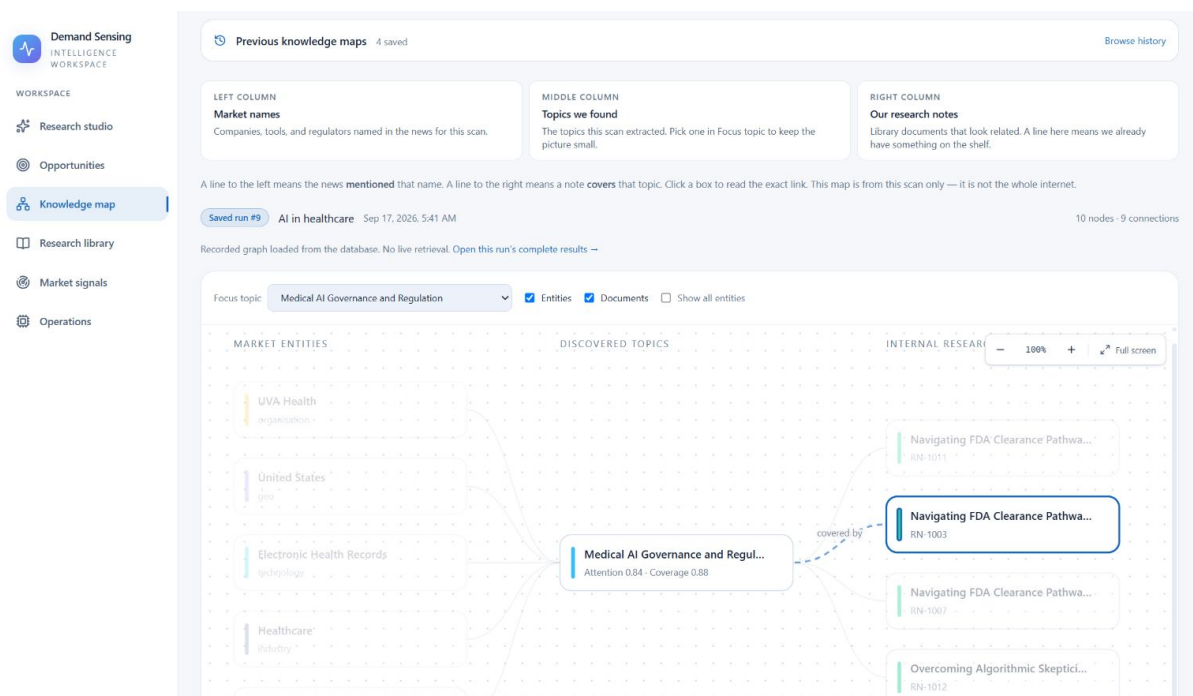
Open a recorded result instantly. No news search and no model calls. Use this first if you want the feel of the product without waiting for a live scan.



Previous runs list in Research studio

Knowledge map

Entities named in the news (left), canonical topic (centre), accepted internal notes (right). A line to the left is a mentions link. A line to the right is covered by. The map is scoped to that scan, not the whole internet.



Knowledge map for Medical AI Governance and Regulation

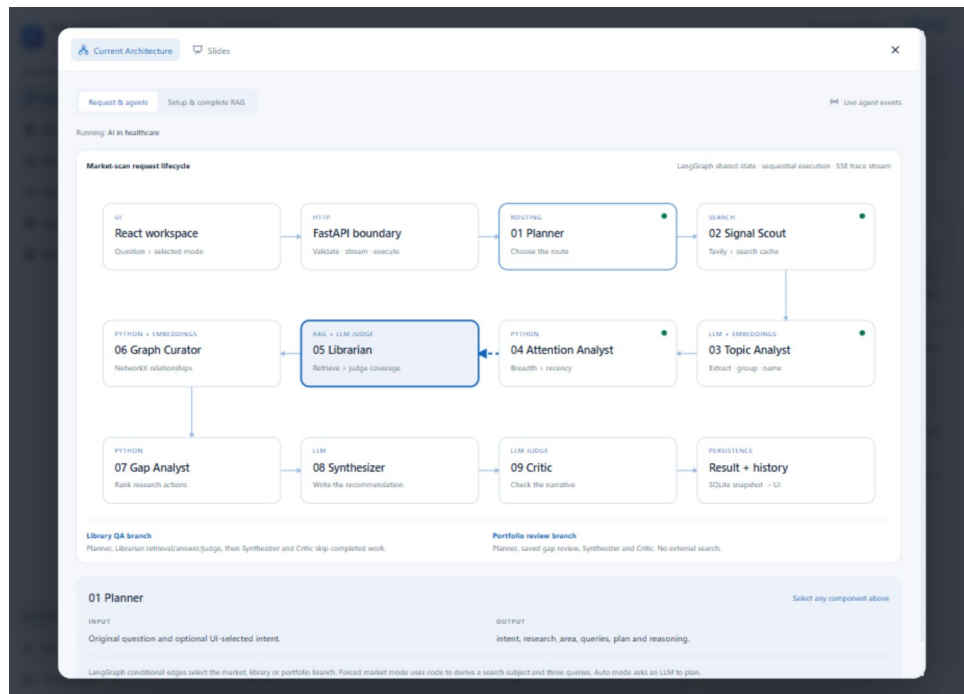
Fullscreen view of the same map:



Fullscreen knowledge map

Current Architecture

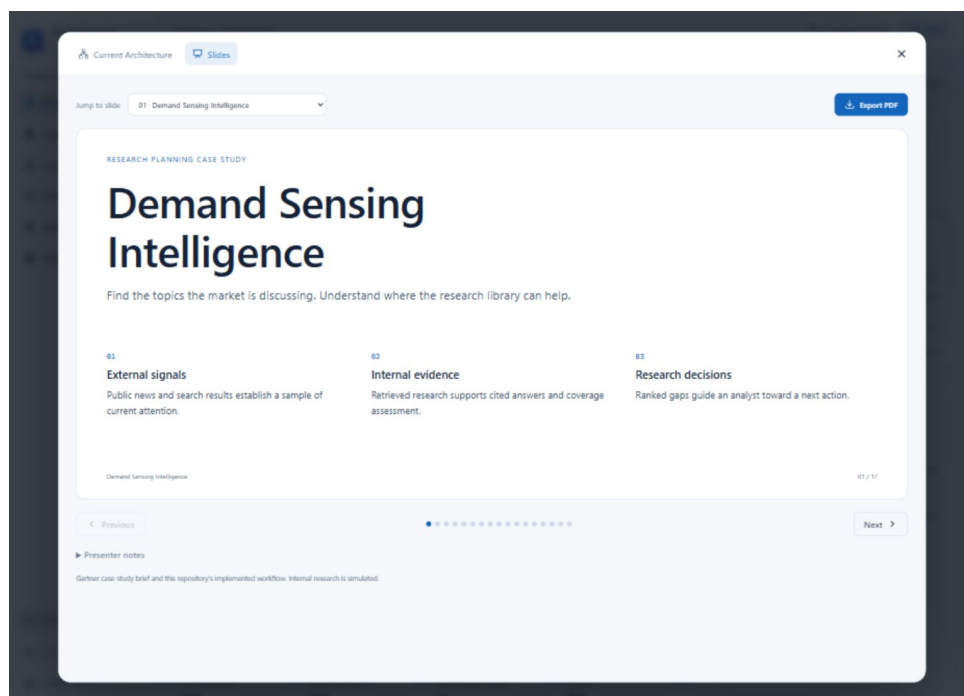
Top-right Current Architecture opens a live-wired map of the LangGraph workflow. Click a node for inputs and outputs.



Current Architecture overlay

Slides

Top-right Slides opens the executive deck (22 pages: 18 main + 4 reference). Download PowerPoint or PDF from that viewer. Those files are presentation/Demand-Sensing-Executive.pptx and presentation/Demand-Sensing-Executive.pdf, and the same copies are served in the app.



Slides overlay

What you need on the machine

- Python 3.11+ (developed on 3.13). Check: `python --version`
- Node.js 20+ (includes npm). Check: `node --version` and `npm --version`
- The `.env` file from email, saved as `backend/.env`

Windows, macOS, and Linux all work. Commands below show PowerShell. On macOS/Linux use `backend/venv/bin/python` instead of `backend\venv\Scripts\python.exe`.

1. Get the code

```
git clone https://github.com/yashugupta786/Agentic-Research-Intelligence-Platform.git
cd Agentic-Research-Intelligence-Platform
```

The repository is public. Anyone with the link can clone it or download the ZIP from GitHub without signing in. API keys are still not in the repo.

Or unzip the archive and open that folder.

Confirm these files exist before installing anything:

```
backend/data/intelligence.db
backend/data/library.faiss
backend/data/library_ids.json
backend/.env.example
```

If the `.db` or `.faiss` file is missing, stop and ask for a complete checkout. Do not run the seed script to “fix” a missing index unless you were told to. Seeding spends Gemini quota and can take a long time.

2. Add API keys (from email)

- 1 Copy `backend/.env.example` to `backend/.env`
- 2 Paste the two keys from the email:

```
GOOGLE_API_KEY=...
TAVILY_API_KEY=...
```

- 1 Save. There should be no quotes around the keys and no `backend/.env` in Git.

The app reads only `backend/.env`. A `.env` in the repo root is ignored.

Keys are required only for a new live scan or a new library answer. If Gemini or Tavily is unavailable at their end, they can still browse the library and open Previous runs. Those screens load from SQLite only. They do not call Gemini, Tavily, or rebuild FAISS.

The committed database already includes complete snapshots from this machine, including:

Run	Mode	Question	What they see
#9	Market scan	AI in healthcare	23 articles, 9 topics, 55-node graph (~64 s)

Run	Mode	Question	What they see
#10	Market scan	AI in supply chain	23 articles, 14 topics, 87-node graph
#11	Market scan	AI in healthcare	23 articles, 10 topics, 63-node graph
#12	Ask the library	What are the risks of AI in healthcare	Cited answer
#13	Portfolio review	same question	Saved gap review

Older rows in Previous runs (ids 1–8) are leftover traces without a full snapshot. Use #9 (or #10 / #11) for the product feel.

3. Backend (one-time)

From the project root:

```
cd backend
python -m venv venv
.\venv\Scripts\python.exe -m pip install --upgrade pip
.\venv\Scripts\python.exe -m pip install -r requirements.txt
```

macOS / Linux:

```
cd backend
python3 -m venv venv
./venv/bin/python -m pip install --upgrade pip
./venv/bin/python -m pip install -r requirements.txt
```

Start the API

Option A — from backend/ (recommended)

```
.\venv\Scripts\python.exe run.py
```

Option B — from the project root (uses backend/venv if it exists)

```
python run.py
```

Leave this terminal open. You should see:

```
Uvicorn running on http://127.0.0.1:8000
Ready | index=435 vectors | llm=True | web_search=True
```

index= should be about 435. That number comes from the committed FAISS file, not from a setup job. `llm=True` and `web_search=True` mean the `.env` keys were picked up. If they are `False`, Previous runs and library browse still work; only **new** questions will fail.

Quick check: <http://127.0.0.1:8000/api/health>

4. Frontend (one-time, second terminal)

Keep the backend running. Open a new terminal at the project root:

```
cd frontend
npm install
npm run dev
```

Open <http://127.0.0.1:5173> — that is the app.

Vite proxies `/api` to port 8000. Always use the 5173 URL in the browser, not 8000, or live agent updates will not work.

5. First ten minutes

- 1 Land on Research studio.
- 2 Open Previous runs and load run #9 (AI in healthcare). This is a full saved scan: topics, coverage, agent trace, and knowledge map. Gemini is not used.
- 3 If keys work, optionally mode Ask the library with what are the major risks of AI adoption in healthcare?. If Gemini does not work, open saved run #12 instead.
- 4 Optional new Market scan with AI in healthcare. That path needs Gemini and Tavily. Skip it if those services fail.

Do not close the tab until a **new** live run finishes if you want that extra result saved. The runs listed above are already in the database you cloned.

Safe demo questions

- Market scan: AI in healthcare
- Market scan: AI in supply chain
- Ask the library: what are the major risks of AI adoption in healthcare?

Workspace map

Place	What it is
Research studio	Ask a question. Market scan, Ask the library, or Auto route. Previous runs.
Opportunities	Saved demand-vs-coverage portfolio (not a second teaching chart of the current scan).
Knowledge map	Query-scoped graph: entities, topics, accepted notes.
Research library	Browse notes and ask RAG questions.
Market signals	Stored Tavily articles from scans.
Operations	Topology and run mechanics.
Current Architecture	Overlay: LangGraph blueprint. Click a node for detail.
Slides	Overlay: executive PowerPoint renders, plus PPTX/PDF download.

Everyday start (after install)

Terminal 1 — from the project root:

```
python run.py
```

(Or `cd backend` then `.\venv\Scripts\python.exe run.py`)

Terminal 2:

```
cd frontend  
npm run dev
```

Browser: <http://127.0.0.1:5173>

What not to do

Don't	Why
Commit or paste <code>.env</code> into Git	Keys
Run <code>python -m scripts.seed</code>	Rebuilds the library; uses quota; not needed
Open the UI on port 8000	SSE proxy is on 5173
Use <code>uvicorn --reload</code>	Can kill a long scan
Treat attention as proven client demand or surge	Attention is from the current news sample, not a historical baseline

Troubleshooting

`llm=False` or `web_search=False` at startup `backend/.env` is missing, in the wrong folder, or the variable names are wrong. Restart `run.py` after saving `.env`.

`index=0` or “library files are missing” `intelligence.db`, `library.faiss`, and `library_ids.json` must sit together in `backend/data/`. Re-clone / re-copy the repo; do not seed unless asked.

Port 8000 already in use Stop the other process, or change the port in `backend/run.py` and `frontend/vite.config.ts` (`proxy.target`) to match.

Port 5173 already in use Vite will pick 5174. Use the URL it prints. Backend CORS allows 5173 by default; stay on 5173 if you can.

Frontend loads, every `/api` call fails Backend is not running, or it is not on `127.0.0.1:8000`.

`npm install` fails Use Node 20+. Delete `frontend/node_modules` and run `npm install` again. A lockfile is included (`package-lock.json`).

`pip install faiss-cpu` fails You need a 64-bit CPython 3.11–3.13. 32-bit Windows Python will not work.

Scan sits still, then errors Usually Gemini or Tavily quota (HTTP 429). Library browse and Previous runs still work without a new scan. Retry after a few minutes.

Windows: python not found Try `py -3.13` or `py -3`. Create the venv with that launcher: `py -3 -m venv venv`.

Optional checks

From backend/ with the venv active:

```
.\venv\Scripts\python.exe -m unittest discover -s tests -v
```

These tests mock providers and do not need live keys.

Project layout

backend/	FastAPI, agents, SQLite, FAISS
run.py	Start the API
.env	Keys (email only – not in Git)
data/intelligence.db	Simulated library + saved runs
data/library.faiss	Chunk index
data/library_ids.json	Vector → document/chunk map
frontend/	React UI
presentation/	Executive PPTX, PDF, slide renders
screenshot/	Product screenshots used in this README
scripts/	Rebuild the executive deck if needed
run.py	Same API start from the repo root
README.pdf	Printable copy of this guide

The in-app Slides overlay and presentation/ contain the executive deck (PowerPoint, PDF and speaker notes). That is the explanation sent by email.

Boundaries (please read)

The research notes are simulated. Live market scans rank sampled news attention, not client demand and not a validated surge. Coverage thresholds are prototype heuristics. This is a local MVP: no production login, and one scan at a time is the intended use.